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Choline chloride and ethylene glycol based deep eutectic solvent (DES) *versus* hydroxyl functionalized room temperature ionic liquids (RTILs): assessing the differences in microscopic behaviour between the DES and RTILs†

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With the aim of understanding the differences in the behavior of deep eutectic solvents (DESs) and room temperature ionic liquids (RTILs) in terms of their structure, dynamics, and intra- and intermolecular interactions, three different ILs and one DES having similar functionalities (hydroxyl) have been investigated by using both ensembled average and single-molecule spectroscopic techniques. Specifically, for this purpose, a choline chloride based DES (ethaline) and three hydroxyl functionalized ILs (1-(2-hydroxyethyl)-3-imidazolium bis(trifluoromethanesulfonyl)imide ([OHEMIM][NTf_2]), *N*-(2-hydroxyl ethyl)-*N*-methylpyrrolidinium bis(trifluoromethanesulfonyl)imide ([OHEMPy][NTf_2]), and *N*-(2-hydroxyethyl)-*N*,*N*-dimethylpropan-1-aminium bis(trifluoromethanesulfonyl)imide ([OHC3CH][NTf_2])) are employed and investigated by EPR, time-resolved fluorescence, NMR and FCS studies. Estimation of polarity through EPR spectroscopy has revealed that the hydroxyl ILs employed in these studies are hyper-polar (close to water) in nature, whereas the polarity of the DES is found to be close to those of aliphatic polyhydroxy-alcohols. Interestingly, both time-resolved fluorescence anisotropy and FCS studies on these systems have suggested that the hydroxyl ILs are more dynamically heterogeneous than the DES. More interestingly, PFG-NMR measurements have indicated that the fluid structure of ethaline is relatively more associated as compared to those of the ILs despite the fact that all the cations have the same hydroxyl functionalities. All these investigations have essentially demonstrated that, despite having similar functionalities, both the DES and hydroxyl ILs employed in the present study exhibit microscopic behaviours that are significantly different from each other, indicating the interplay of various intermolecular interactions within the constituent species in governing the behaviours of these solvent systems.

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1. Introduction

The recent upsurge in the interest in room temperature ionic liquids (RTILs) towards both industry and academia stems from their desirable physicochemical properties such as their high chemical and thermal stability, nonflammability, high conductivity, wide liquid ranges and good solubilizing capacity of a number of organic compounds.[1–3] However, non-biodegradability and toxicity are some of the challenging issues which are often encountered in RTILs.[4,5] In contrast to this, deep eutectic solvents (DESs) have emerged as potential substitutes for both RTILs and common volatile organic solvents owing to their biodegradability, biocompatibility, relatively less toxicity and environment friendly nature.[6–9] As both RTILs and DESs share many characteristic physicochemical properties, DESs are often regarded as a subclass of ionic liquids.[8,9] However, researchers have pointed out that despite their similarities both DESs and RTILs are by far different groups of substances.[10,11] In RTILs, apart from some non-covalent interactions (van der Waals interaction, dipole–dipole interaction, *etc.*),

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† Electronic supplementary information (ESI) available: The processes of synthesis and characterization and water contents of the DES and hydroxyl ILs; absorption, emission and fluorescence intensity time traces for neat solvents as well as with probe molecules; Kamlet–Taft parameters for ethaline and [OHC3CH][NTf_2]; anisotropy traces and FCS curves of C153 and MPTS in the DES and hydroxyl ILs; rotational reorientation times of the probe molecules and hydrodynamic radii of the cationic species. See DOI: 10.1039/d1cp05010a

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coulombic interaction between the cations and anions is the dominant contributor to the structures and properties of the ILs.[12,13] However, in the case of DESs, it is believed that complex hydrogen bonding between the constituent species is responsible for governing the macroscopic physicochemical properties of DESs.[7–9] Due to this it has also been pointed out that there are much more differences than similarities between these two classes of systems.[10,11,14] In this scenario, it would be interesting to know the comparative behavior of both DESs and ILs having similar functionalities (hydroxyl group) so as to understand whether the microscopic behavior of these classes of solvent systems is dictated by hydrogen bonding interaction or other types of intermolecular interactions.

RTILs are salts with low melting points which entirely consist of ions and are liquids at room temperature.[1,2,12] Since there are numerous possible combinations of cations and anions, many ILs can be prepared as per the need of the end-user, and because of this, these solvents are termed 'designer solvents'.[2,12] With increase in chemical variety among ILs, they are further classified into task specific ILs, functionalized ILs, protic ILs and poly-ionic ILs.[15–17] In contrast to ILs, DESs are usually prepared by the combination of hydrogen bond donors (HBD) and hydrogen bond acceptors (HBA), which results in a significant decrease in the freezing point temperature owing to the extensive interspecies hydrogen bonding network.[7–9,18] The most common HBA are usually ammonium and phosphonium salts, whereas HBD are usually alcohols, amides and acid functionalities.[7–9] Since the starting materials for the preparation of DESs are relatively cheaper as compared to ILs, large scale preparation of DESs is possible for their use in various fields such as electrochemistry,[19,20] synthesis,[21] catalysis,[18,22] nanotechnology,[23,24] and gas adsorption.[25]

Some notable experimental techniques like fluorescence,[26–32] small angle neutron scattering,[33,34] NMR diffusion[35–39] and theoretical studies based on molecular dynamics (MD) simulation[40–45] have been used to explore the microscopic structural organisation and dynamics of DESs and RTILs independently. Atkins and co-workers[17] suggested that ILs have huge structural diversity depending on the nature of ions as well as their self-organisation into ion pairs, ions' crystal lattice, H-bonding network and micellar domains. Fayer and co-workers[46] also reported that the intra- and intermolecular interactions in ILs have significant impact on the heterogeneous microstructural organisation and liquid morphology which contribute to the dynamic heterogeneity of ILs. Moreover, the complex solvation processes of ILs have also been reported by many groups and it has been demonstrated that the multiple time scales of solvent relaxation phenomena are significantly affected by the heterogeneity of the medium.[47–49] However, for DESs the assessment of the heterogeneity of the medium is not straightforward. For example, Biswas and co-workers[32,50,51] have reported that the non-ionic DES formed by the combination of acetamide and urea is not heterogeneous, whereas the choline chloride/urea combination or acetamide/electrolyte mixture are both spatially and dynamically heterogenous. Furthermore, the DES ethaline, a mixture of choline chloride and ethylene glycol, is reported to be spatially homogeneous but dynamically heterogeneous.[52,53] Though there are a few studies which describe the structural organisation of some neat ILs and DESs, the understanding about the differences in microscopic structural organisation between these two classes of substances still remains hazy. Therefore, it would be interesting to probe the differences in microscopic behaviour between DESs and ILs in terms of the nature of solute–solvent interaction, intermolecular interaction and dynamics of the solvent systems by employing DESs and ILs having similar functional moieties. Studies on this aspect are expected to be helpful in understanding the structure–property relationship for these classes of substances (DESs and RTILs).

In light of the foregoing discussion, the current study focuses on the understanding of differences in microscopic behaviour between DESs and ionic liquids. For this purpose, a DES, namely ethaline (choline chloride + ethylene glycol 1 : 2 ratio), and three other hydroxyl group containing room temperature ionic liquids are employed for the study. The hydroxyl group based ILs have been chosen purposefully so that apart from the role of the hydroxyl group, various other factors which account for the microscopic behaviour of both DES and ILs can be clearly understood. Furthermore, for a fair comparison between the DES and ionic liquids, a cholinium ionic liquid ([OHC3CH][NTf_2]) (Chart 1) consisting of the propyl chain has also been used for this study. The differences in microscopic behaviour between the DES and hydroxyl ILs have been investigated with the help of different spectroscopic techniques such as steady state and time resolved fluorescence spectroscopy, fluorescence correlation spectroscopy (FCS), electron paramagnetic resonance (EPR) spectroscopy and nuclear magnetic resonance (NMR) spectroscopy so that information regarding the local environment, the solute–solvent interaction and the dynamics of the solvent systems can be obtained. Especially, the rotational and translational diffusion dynamics of some selected probes determined through time resolved fluorescence anisotropy measurement and FCS techniques, respectively,

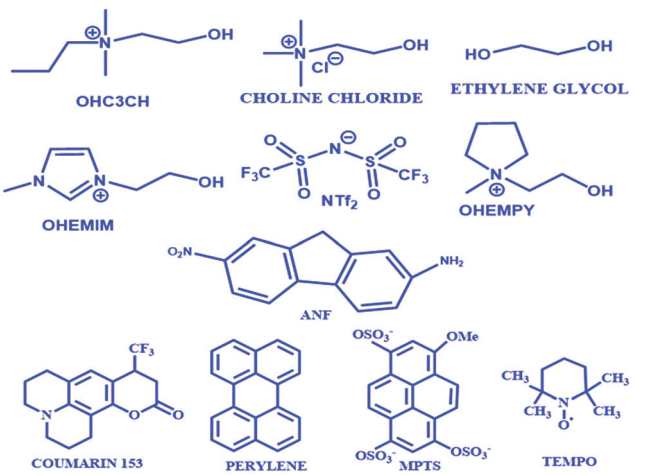


Chart 1 Molecular structures of the components of the DES, hydroxyl ILs and different probes.

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have been used to differentiate the local environment and nature of solute–solvent interaction between the DES and ILs. Moreover, the polarities and self-diffusion coefficients of the concerned solvent systems have been estimated by exploiting EPR and NMR techniques respectively. All these investigations have revealed significant differences in microscopic behaviours between two ILs and the DES despite the fact that they have the same (hydroxyl) functionalities. This observation indicates that in addition to intermolecular hydrogen bonding interaction the interplay of various other types of intermolecular interactions within the constituent species can also have significant impact in governing the behaviours of these solvent systems.

2. Experimental

2.1. Materials

Ethylene glycol, 1-methylimidazole, *N*-methylpyrrolidine, *N*,*N*-dimethylethanolamine, 1-bromopropane, Coumarin (153), perylene, 8-methoxypyrene-1,3,6-trisulfonic acid trisodium salt (MPTS), and 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO) were purchased from Sigma-Aldrich. Choline chloride and lithium bis(trifluoromethanesulfonyl)imide ($LiNTf_2$) were received from TCI. 1-(2-Hydroxyethyl)-3-imidazolium bis(trifluoromethanesulfonyl)imide was purchased from io-li-tec. 2-Chloroethanol was purchased from HiMedia. All the chemicals were dried under vacuum and used without any further purification. DMSO-d^6 from Sigma Aldrich was used as solvent for recording NMR. The molecular structures of the components of the DES, hydroxyl ILs and different probe molecules used in this study are provided in Chart 1. The details regarding the synthesis, characterization, water content and sample preparation are provided in the ESI.†

2.2. Instrumentation

The absorption spectra of the samples were collected using a Cary 100 biospectrophotometer. A Cary Eclipse fluorescence (Agilent Technologies) spectrophotometer was used for recording the steady state fluorescence spectra of the samples. A time correlated single photon counting (TCSPC) spectrometer (OB 920) was used for the measurements of time resolved fluorescence anisotropy with the help of a 405 nm picosecond diode laser (EPL) as an excitation source. A dilute solution of Ludox in water was taken as a scatterer to obtain the instrument response function (IRF) and it was found to be around 80 ps for the 405 nm laser. Fluorescence intensities for parallel ($\parallel$) and perpendicular ($\perp$) polarization were collected alternatively until a peak difference between the parallel ($\parallel$) and perpendicular ($\perp$) decays of $\sim$5000 (at t = 0) was observed. The same procedure was applied for G-factor calculation but with horizontal polarization of the exciting laser beam and 5 cycles of repetition. The temperature of the cell was monitored by circulating water through the cell holder using a Quantum along with a North West (TC 125) temperature controller. A Hamamatsu microchannel plate photomultiplier tube (R3809U-50) detector was used to collect the decay signals of the samples. The decay curves were analysed with the help of F980 decay analysis software. The decay curves were fitted through non-linear least squares iteration methods and the quality of the fitting was judged from the chi-square (χ^2) values as well as by verifying the weight deviation. The viscosities of all the samples were measured using an LVDV-III Ultra Brookfield Cone and Plate viscometer (experimental error: $\pm$2%). A Bruker Super QCW-EPR bridge (model A300) X band spectrometer was used for recording the EPR spectra and WinEPR SymFonia software (version 1.26 beta) was used for simulation of the EPR spectra. All the NMR measurements were carried out on a 9.4 T Bruker Avance NMR spectrometer with 400.1 MHz Larmor frequencies for H^1.

For FCS measurements, a time-resolved confocal fluorescence setup (MicroTime 200, PicoQuant), equipped with an inverted microscope (Olympus IX 71) along with a water immersion objective (Olympus UPlanSApo NA 1.2, 60×), was used. The excitation source was a pulsed diode laser (403 nm) with an FWHM of 176 ps and a repetition rate of 20 MHz. The laser beam was made to pass through a polarization maintaining single-mode optical fiber and focused onto the sample (which was placed on a coverslip) using a water immersion objective which was also used to collect the fluorescence light emitted from the samples. The emitted beam was then filtered by using a 425 nm filter to remove any excitation light. The signal was then focused onto a 50 μm diameter pinhole to remove the out-of-focus signal, recollimated, and directed onto a (50/50) beam splitter prior to entering two single-photon avalanche photodiodes (SPADs). The signals from the two SPAD detectors were cross-correlated to generate fluorescence correlation traces. A SymPhoTime software controlled PicoHarp300 TCSPC module was used for data acquisition in a time-tagged time-resolved (TTTR) mode. All FCS experiments were carried out at room temperature ($\sim$298 K). During all FCS measurements, the excitation power was kept at $\sim$4 μW. The excitation power was kept low during all FCS studies so as to avoid the contribution of background noise. The absorption and emission spectra as well as the plots of fluorescence intensity time traces for both neat solvents and the probe molecules are provided in Fig. S1 and S2 (ESI†) to demonstrate the optical cleanness of the concerned media.

2.3. Methods

The time resolved fluorescence anisotropy was calculated by using the following equation:[54]

$$r(t) = \frac{I_{VV}(t) - GI_{VH}(t)}{I_{VV}(t) + 2GI_{VH}(t)} \quad (1)$$

where I_{VV} is the fluorescence intensity with parallel polarisation, *i.e.* vertically polarised excitation with vertically polarized emission, and I_{VH} is the fluorescence intensity with perpendicular polarisation, *i.e.* vertically polarized excitation with horizontally polarized emission. G is the instrument correction factor for the detector sensitivity to the polarization of the

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emission. It is given by the equation

$$G = \frac{\sum I_{HH}(t)}{\sum I_{HV}(t)} \quad (2)$$

where $\sum$ is the summation of all fluorescence intensities (in our case it is the sum of the fluorescence intensities for the five cycles), I_{HH} is the fluorescence intensity with horizontally polarized excitation and horizontally polarized emission and I_{HV} is the fluorescence intensity with horizontally polarized excitation and vertically polarized emission. The anisotropy data were fitted to calculate the rotational relaxation time by the following equation:

$$r(t) = r_0 e^{\left(-\frac{t}{\tau_r}\right)} \quad (3)$$

where r_0 is the initial anisotropy and τ_r is the rotational relaxation time.

The self-diffusion coefficients of all the samples were determined using pulse field gradient (PFG) NMR techniques. A stimulated echo bipolar pulse-gradient pulse (stebpgp) sequence was applied with a maximum gradient pulse strength of 50 G cm^{-1}. The echo heights were attained by varying the gradient pulse strength from 2 to 95% at 16 equal intervals. The echo heights were fitted using the Stejskal–Tanner equation,[55] which is given below:

$$S(g) = S(0)\exp -D\gamma^2\delta^2 g^2\left(\Delta - \frac{\delta}{3}\right) \quad (4)$$

where $S(g)$ and $S(0)$ are the echo heights at the gradient strengths g and 0, respectively. γ and δ are the gyromagnetic ratio of the proton and gradient pulse length, respectively, while Δ is the duration between the two gradient pulses. The experimental error limit for the PFG NMR diffusion experiment is $\sim\pm10\%$.

The FCS technique is a very sensitive and powerful tool for estimating the translational diffusion (D) of a probe molecule in the concerned solvent system. This measurement was performed by placing a highly diluted sample (10–20 nM) over a coverslip which in turn was placed over the water immersion objective through which the excitation laser light was passed. The fluctuations in fluorescence intensity of the sample were measured in very small observation volumes ($\sim$1 fL) with the help of a pinhole. Now, these fluctuating fluorescence signals were detected using two SPAD detectors and were cross-correlated to generate the correlation curves $G(\tau)$. The decay of the correlation function with respect to time was used for obtaining dynamical information on translational diffusion. The correlation function $G(\tau)$ of the fluorescence intensity is given by[54]

$$G(\tau) = \frac{\langle \delta F(t)\delta F(t+\tau)\rangle}{\langle F(t)^2\rangle} \quad (5)$$

where $\delta F(t)$ and $\delta F(t+\tau)$ represent the fluctuations in fluorescence intensity around the mean value at times t and $(t+\tau)$, respectively, and $F(t)$ is the average fluorescence intensity. The correlation data were fitted to both single-component diffusion ($\beta = 1$) and anomalous diffusion ($\beta < 1$) using the equation[56]

$$G(\tau) = \left[\frac{1}{N \times \left[1 + (\tau/\tau_D)^{\beta}\right] \times \left[\sqrt{1 + (\tau/\tau_D)^{\beta}\left(\frac{1}{k^2}\right)}\right]}\right] \quad (6)$$

In eqn (6), N represents the average number of fluorescent molecules in the observation volume, τ is the delay or lag time, and τ_D is the translational diffusion time constant or the time taken for the fluorophore molecule to diffuse through the observation volume. β ($0 < \beta < 1$) is the stretching exponent representing the distribution of τ_D and k represents the aspect ratio of the observation volume which is given by $k = z/w_{xy}$, where z and w_{xy} are the longitudinal and transverse radii of the observation volume, respectively. The excitation volume of the FCS setup was calibrated by taking a dilute solution of Rhodamine 6G (R6G) in water whose translational diffusion coefficient (D_t) is 426 $\mu m^2\ s^{-1}$.[57] The estimated excitation volume was $\sim$0.45 fL for 405 nm laser sources. The D_t value was calculated by using eqn (7) which is provided below:

$$D_t = \frac{{w_{xy}}^2}{4\tau_D} \quad (7)$$

3. Results and discussion

3.1. Steady state spectral measurements

The steady state absorption and emission spectra of a dipolar probe, C153, in both the DES and hydroxyl functionalized ILs (hydroxyl ILs) are provided in Fig. 1. The absorption and emission maxima of C153 in all the concerned solvents are given in Table 1. From Fig. 1, one can observe that there are hardly any differences observed in the emission or absorption spectra of C153 in the DES and hydroxyl ILs. From Table 1 also no noticeable changes in absorption or emission maxima of C153 are observed in all the solvent systems. So, under the current experimental conditions and looking only at their steady state spectra, both the DES and hydroxyl ILs seem to be indistinguishable from each other, indicating similar polar environments for both the DES and hydroxyl ILs.

However, just by monitoring the absorption and emission spectra of C153 in the concerned solvent, it is often difficult to comprehend the medium polarity accurately. This is because the difference between the ground state and excited state dipole moments is not very high for C153 ($\sim$6 Debye).[58,59] Moreover, effects such as solute–solvent hydrogen bonding interaction are not well accounted for C153 when the polarity of a solvent is estimated through steady state spectral shift of C153.[58,59] Here, we would also like to note that many previous studies[29–32,59,60] on DESs and ionic liquids also fail to distinguish completely the medium polarity only by observing the steady state spectral properties of C153. Since the polarity of the medium is often

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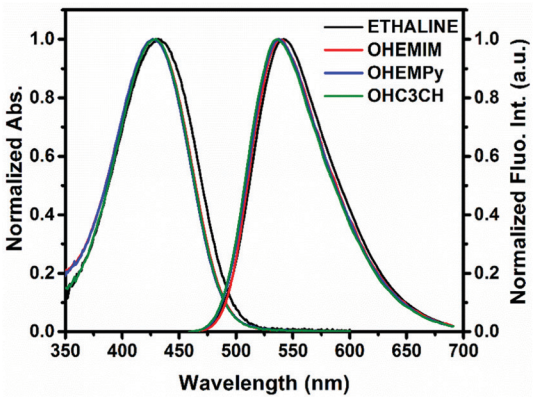


Fig. 1 Normalized absorption (left side) and emission spectra (right side) of C153 in ethaline and the hydroxyl ILs. λ_{ex} = 405 nm.

Table 1 Absorption and emission maxima of C153 in various solvent systems

System	λ_{max}(abs) (nm)	λ_{max}(ems) (nm)
Ethaline	431	539
[OHEMIM][NTf_2]	429	538
[OHEMPy][NTf_2]	428	536
[OHC3CH][NTf_2]	428	537

estimated spectroscopically by measuring the transition energy of Reichardt's dye 30 in that solvent, we have also made an attempt to estimate the polarity of the media that are used in this study through this method.[59,61] The absorption spectra of Reichardt's dye 30 for the current solvent systems (DES and hydroxyl ILs) are provided in Fig. S3 (ESI†). From Fig. S3 (ESI†) it can be seen that the long wavelength CT band in [OHEMIM][NTf_2] and [OHEMPy][NTf_2] is missing due to which it is not possible to estimate the solvent polarity parameter (E_T(30)) and the Kamlet–Taft parameter for all the solvents (Table S2, ESI†). It is not surprising as Reichardt's dye 30 is found to be insoluble in many molecular liquids, and more importantly, the long wavelength CT band also remains absent in many molecular solvents.[61] The absence of the long wavelength CT band of Reichardt's dye 30 can be explained by considering the protonation of phenolate oxygen atoms arising from the acidity of these hydroxyl ILs.[61]

Considering the above discussion, we have employed electron paramagnetic resonance (EPR) spectroscopy which proves to be an important tool and alternative to the aforementioned methods for estimating the small change in medium polarity precisely by monitoring the EPR spectra of free radical probes such as 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO) in media like ionic liquids.[31,62,63] The nitroxide radical of TEMPO usually exists in a charge transfer (CT) structure and shows a triplet hyperfine structure with unequal intensities in the EPR spectra.[62–64] The nitrogen isotropic hyperfine coupling constant is usually described by the parameter (a_N/G) with units of Gauss. The (a_N/G) value usually represents the measure for perturbations of the spin-density distribution of the radicals (in our case it is TEMPO) in solution caused by electrostatic interactions and hydrogen bonding with the solvent. It has also been reported that the value of (a_N/G) depends on the spin density around the nitrogen atom. With increase in polarity of the medium, the stabilization of the ground state CT structure of the nitroxide radical increases the spin density around the nitrogen atom, which subsequently increases the (a_N/G) value of TEMPO. Thus, a quantitative idea about the difference in polarity between the DES and hydroxyl ILs can be obtained by estimating the a_N/G value of TEMPO in the concerned solvents. The EPR spectra of TEMPO, along with the corresponding simulated spectra, in both the DES and hydroxyl ILs are shown in Fig. 2. From Fig. 2, one can notice the difference in the line shapes of the EPR signals in all the media. Here we would like to mention that the line shape of EPR spectra usually reflects the spin relaxation process due to the tumbling motion of TEMPO radicals.[62,63] EPR spectra that differ in line shape indicate the difference in the rotational behaviour of TEMPO in these solvents.[62–64] It has been reported earlier that the rotational motion of TEMPO in a particular solvent is significantly affected by the viscosity, intermolecular hydrogen bonding interaction of TEMPO with the solvent molecule and extent of microheterogeneity of the solvent.[63,64] Since the aforementioned properties of the concerned solvent systems are different for each solvent, it is reasonable to observe a difference in the line shape of the EPR spectral signal due to a change in the microenvironment of these solvent systems.

Here we would like to note that for viscous and confined media like DESs and RTILs, the assessment of a_N/G values from the maxima of two successive hyperfine lines may lead to imprecise estimation of the a_N/G value because of spectral broadening.[64] In such a scenario, the a_N/G values have been estimated from the centre point of two consecutive hyperfine lines. Moreover, EPR spectra of TEMPO dissolved in a series of known solvents of varying polarities have also been recorded and their corresponding a_N/G values have been estimated with

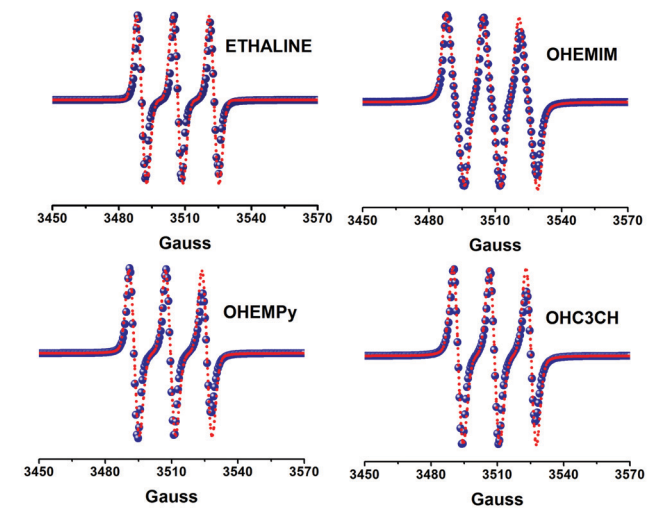


Fig. 2 EPR spectra of TEMPO (blue spheres) in the DES and hydroxyl ILs. The red lines represent the corresponding simulated spectra.

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Table 2 The a_N/G values of TEMPO in DESs and hydroxyl ILs and their estimated $E_T(30)$ values[a]

Solvent	a_N/G	$E_T(30)$/kcal mol^{-1}
Ethaline	16.31 ± 0.012	58.6 ± 0.015
[OHEMIM][NTf$_2$]	16.57 ± 0.011	61.8 ± 0.017
[OHEMPy][NTf$_2$]	16.49 ± 0.012	60.4 ± 0.015
[OHC3CH][NTf$_2$]	16.52 ± 0.018	60.8 ± 0.017
[BMIM][NTf$_2$][a]	16.03 ± 0.019	52.5 ± 0.02
Ethylene glycol	16.24 ± 0.017	56.3 ± 0.015
1,4-Dioxane	14.6 ± 0.015	36.0 ± 0.017
Chloroform	14.87 ± 0.016	39.1 ± 0.018
Acetone	15.04 ± 0.018	42.2 ± 0.015
Ethyl alcohol	15.82 ± 0.017	51.9 ± 0.016

[a] The value is taken from ref. 60.

the help of spectral data. In addition to this, the solvent polarity parameters ($E_T(30)$) of these known solvents are collected from the literature.[65] Then a calibration curve is generated by plotting the a_N/G values (Y-axis) of TEMPO in these known organic solvent against their corresponding $E_T(30)$ values (X-axis). The $E_T(30)$ values for various known solvents are provided in Table 2. Fig. 3 presents the plot of a_N/G *versus* $E_T(30)$ for various solvent systems. Now by estimating the a_N/G values for unknown solvent systems and then by using the calibration curve, the $E_T(30)$ values of the unknown solvent systems are determined.

From Table 2, it can be noticed that the $E_T(30)$ values of the hydroxyl ILs are found to be higher than that of ethaline, indicating that the micro-polarities of the hydroxyls ILs are relatively higher than that of ethaline. More interestingly, the $E_T(30)$ values of all the hydroxyl ILs are found to be close to that of water ($E_T(30)$ = 63.1 kcal mol^{-1}).[65] Deng and co-workers,[61] while determining the polarity of some non-hydroxyl and hydroxyl functionalized ILs through solvatochromic studies of Reichardt's dye, also found the unusual hyper-polarity of hydroxyl ILs. Through computational studies they demonstrated that the intramolecular synergistic effect that exists between the hydroxyl group of the cation and the corresponding anion is

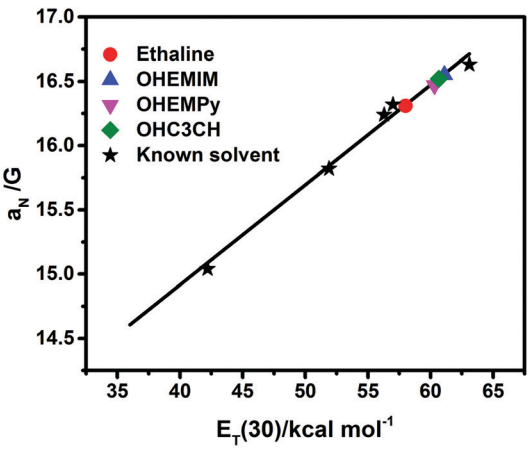


Fig. 3 Plot of the a_N/G *vs.* $E_T(30)$ values of TEMPO in both the DES and hydroxyl ILs. The black stars indicates the $E_T(30)$ values corresponding to known solvents.

primarily responsible for the unusual hyper-polarity of hydroxyl ILs. Furthermore, Shibuya and co-workers[62] suggested that the oxygen of ATEMPO (4-amino-2,2,6,6-tetramethylpiperidine-1-oxyl) can interact with the C2 hydrogen of 1-butyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide through hydrogen bonding interaction and thus increase the a_N/G and polarity values of the medium. In light of these findings, in the present study, we can infer that in [OHEMIM][NTf$_2$] both C2 hydrogen and the hydroxyl group can form H-bonding interaction with the anion as well as with the oxygen of the nitroxide radical and thus enhance the polarity of the medium to a greater extent as compared to those of the other two hydroxyl ILs. Here we would like to mention that the hyper-polarity of the hydroxyl ILs is one of the interesting findings as these functionalized ILs can be potential candidates for highly polar non-acidic and non-aqueous media which can be used as substitutes for water and highly polar organic solvents in synthesis and catalysis.[61]

As both the DES and hydroxyl ILs are viscous media, where solvation of a solute fluorophore is relatively slow as compared to common solvents such as water, DMSO, and ethanol, fluorescence can be observed from different unrelaxed states of the solvent around the fluorophore in these media.[51,58] Especially, when the fluorescence lifetime of the fluorophore is much shorter than the average solvation time in a particular solvent, fluorophores may exist in a state where they are not fully solvent-equilibrated. Under this condition the micro-environments around the fluorophores are different and they show a shift in the emission maxima upon varying the wavelength of the excitation source. This is a very well-known method for determining the micro-heterogeneous behaviour of a given heterogeneous solvent.[51,58,66,67] In the current studies, we have tried to explore the difference in the micro-heterogeneous behaviour of the DES and hydroxyl ILs by employing excitation wavelength (λ_{ex}) dependent shift in fluorescence maxima (λ_{em}^{max}) of 2-amino-7-nitrofluorene (ANF). ANF has a lifetime of nearly 50 picoseconds and it is known to exhibit shift in λ_{em}^{max} by increasing the λ_{ex} towards longer wavelengths.[31,66,67] Upon looking at Fig. 4, a shift of 8 nm of λ_{em}^{max} is observed in the case of ethaline by increasing the λ_{ex} from 390 to 510 nm. However, the shift in λ_{em}^{max} is found to be more than double in the case of [OHEMPy][NTf$_2$], indicating more spatial heterogeneity in the medium as compared to ethaline. Furthermore, by comparing ethaline with its IL analogue ([OHC3CH][NTf$_2$]), the observed shift in λ_{em}^{max} is also found to be greater (15 nm) for the later than the former. This observation indicates that the DES under study is spatially less heterogeneous as compared to the hydroxyl ILs. Here we would like to mention that ANF also shows a shift of 8 nm of the absorption maxima upon going from [OHEMIM][NTf$_2$] to ethaline (Fig. S4(a), ESI†), which is because of the difference in the medium polarity. However, it has been reported earlier that the extent of shift in λ_{em}^{max} with increase in λ_{ex} is not because of the shift in λ_{max}^{abs} but arises because of the inhomogeneous broadening of the absorption spectra due to the distribution of the solute–solvent interaction energy in these confined media which is found to be more for the hydroxyl ILs (FWHM = 112 nm) as compared to

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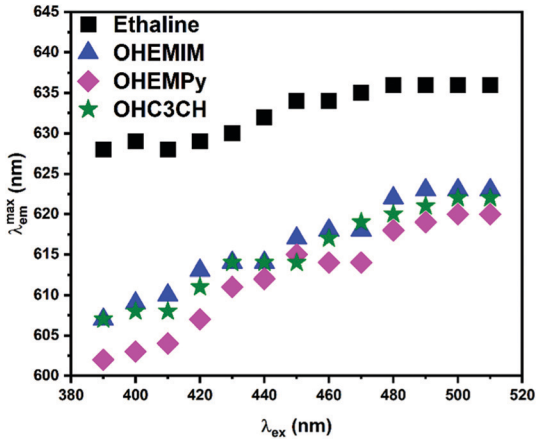


Fig. 4 Plots of λ_{em}^{max} *versus* λ_{ex} of ANF at 298 K of the DES and hydroxyl ILs (experimental error = ±2 nm).

the DES (FWHM = 101 nm).[58,67] Moreover, the average lifetimes (τ_{avg}) for ANF in both ethaline and the hydroxyl ILs are found to be comparable ($\tau_{avg} \leq 40$ ps) (Fig. S4(b), ESI†). But the average solvation times (τ_{solv}) in these media, as reported earlier (*e.g.* τ_{solv} = 160 ps for ethaline and 560 ps for OHEMIM),[31,68] are found to be much higher than the τ_{avg} of ANF. Therefore, it is possible to observe fluorescence from the excited species which are not fully solvent equilibrated.[58,67] As shown before the τ_{solv} values are estimated to be different for different solvents. Hence, as the average solvation times (as well as viscosities) in the hydroxyl ILs are much higher as compared to that of ethaline, it is reasonable to expect more shift in λ_{em}^{max} with increase in (λ_{ex}) for the former than the later. Though the investigation of λ_{ex} dependence shift in λ_{em}^{max} cannot provide the exact length scale at which the particles are correlated, it only provides a qualitative idea about the medium heterogeneity.

3.2. Rotational dynamics through time resolved fluorescence anisotropy measurement

Since DESs also differ from ILs based on their constituent materials and their preparation method, the local structure organisation of DESs could be different from that of ILs. So, to know more about the difference in microscopic behaviour in terms of solute–solvent interactions in the given micro-heterogeneous systems like the DES and hydroxyl ILs, rotational dynamics for three probe molecules C153 (dipolar), perylene (apolar) and MPTS (charged) with different chemical characteristics have been investigated through time-resolved fluorescence anisotropy measurement.

3.2.1. C153. Fig. 5a presents the plot of time resolved fluorescence anisotropy decay for C153 in both the DES and hydroxyl ILs at 298 K. The data have been fitted to a single exponential decay function to obtain the rotational reorientation time (τ_r) of C153 in both the DES and hydroxyl ILs. The measured τ_r values along with viscosity at 298 K are given in Table 3, while for other temperatures the data are provided in Table S3 (ESI†). From Fig. 5a, a faster anisotropy decay of C153 in ethaline as compared to the hydroxyl ILs is observed owing to the lower bulk viscosity of the former than the later. From Table 3, one can also find that the experimentally observed τ_r value of C153 in ethaline is relatively lower as compared to those in the hydroxyl based ILs. Again, with increase in temperature, a gradual decrease in the τ_r value can be observed in all the solvent systems, which is in accordance with the decrease in the bulk viscosity of the medium with increase in temperature (Table S3, ESI†). However, when the data are compared for ethaline and its IL analogue, *i.e.* [OHC3CH][NTf_2], it can be noticed that the τ_r value does not vary proportionately with increase in the value of the viscosity of the medium concerned. For example, upon going from ethaline to [OHC3CH][NTf_2] the viscosity increases by nearly 2.5 times, whereas the increase in τ_r value is estimated to be nearly 2 times. Additionally, we have also measured the τ_r values of C153 in all the concerned media under iso-viscous conditions (Table 4). From Table 4 one can also notice that the rotational diffusion of C153 in the hydroxyl ILs is found to be faster than that in ethaline. All these observations essentially indicate that C153 experiences different frictions in ethaline and [OHC3CH][NTf_2] and the micro-viscosities felt by C153 are different in the two media.

To shed more light on the solute–solvent interaction in the media under study, the τ_r values obtained from time resolved anisotropy data have been analysed by the Stokes–Einstein–Debye

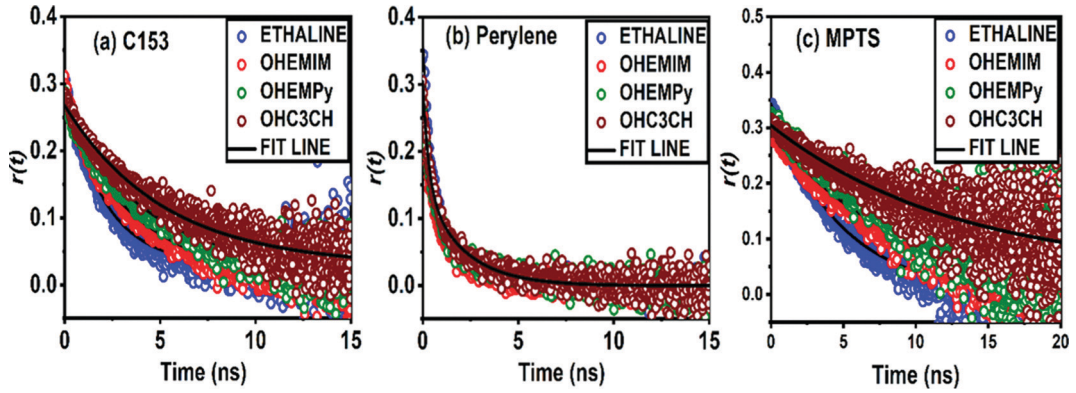


Fig. 5 Time-resolved fluorescence anisotropy decay curves for (a) C153, (b) perylene and (c) MPTS in the DES and hydroxyl ILs at 298 K.

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Table 3 Rotational relaxation parameters of different probes in both the DES and hydroxyl ILs at 298 K

	C153		Perylene		MPTS	
System/η (cP)	τ_r (ns)	C_{obs} (avg.)	τ_r (ns)	C_{obs} (avg.)	τ_r (ns)	C_{obs} (avg.)
Neat ethaline/(38.5)	2.44 ± 0.1	0.65 ± 0.03	0.95 ± 0.04	0.30 ± 0.01	4.79 ± 0.2	1.27 ± 0.06
[OHEMIM][NTf_2]/(72.7)	4.02 ± 0.15	0.61 ± 0.03	1.53 ± 0.06	0.25 ± 0.01	7.12 ± 0.27	1.06 ± 0.05
[OHEMPy][NTf_2]/(92.4)	4.30 ± 0.15	0.55 ± 0.02	1.69 ± 0.06	0.22 ± 0.01	7.64 ± 0.29	0.95 ± 0.05
[OHC3CH][NTf_2]/(98.5)	4.76 ± 0.18	0.55 ± 0.02	1.81 ± 0.06	0.21 ± 0.01	8.91 ± 0.32	1.07 ± 0.05

Table 4 Rotational relaxation parameters of different probes in both the DES and hydroxyl ILs under iso-viscous conditions

System	C153	Perylene	MPTS
Ethaline	2.44	0.95	4.79
[OHEMIM][NTf_2]	2.36	0.94	4.45
[OHEMPy][NTf_2]	2.20	0.81	4.47
[OHC3CH][NTf_2]	2.19	0.82	4.57

equation. SED theory considered the solvent as a structureless continuum, where the rotational relaxation time constant (τ_r) of a medium size solute is related to the bulk viscosity (η) of the medium by the following equation:[54]

$$\tau_r = \frac{\eta v f c}{kT} \quad (8)$$

where k represents the Boltzmann constant, T is the absolute temperature, V is the molecular volume of the solute molecule calculated from van der Waals radii and f is the shape factor of the solute probe. C is the boundary condition parameter having two limiting values, stick ($C = 1$) and slip ($0 < C < 1$). The value of C usually signifies the extent of solute–solvent coupling for a given solute and solvent. The rotational relaxation parameters of the different probe molecules used for the study have been reported in our earlier studies.[31]

From the SED plots of C153 (Fig. 6a) in both the DES and hydroxyl ILs, it can be noted that the values of τ_r of C153 in all the solvent systems lie between the stick and slip boundary lines as predicted by the hydrodynamic theory. A more detailed understanding about the interaction of C153 with both the DES and ILs can be obtained by considering the friction coefficient, C_{obs} value (ratio between experimentally measured τ_r and theoretically calculated τ_r values).[26,68] Since the C_{obs} value is obtained by the aforementioned ratio, the value of C_{obs} usually depicts the extent of solute–solvent interaction.[26,69] Here, we would like to mention that an increase in temperature can also

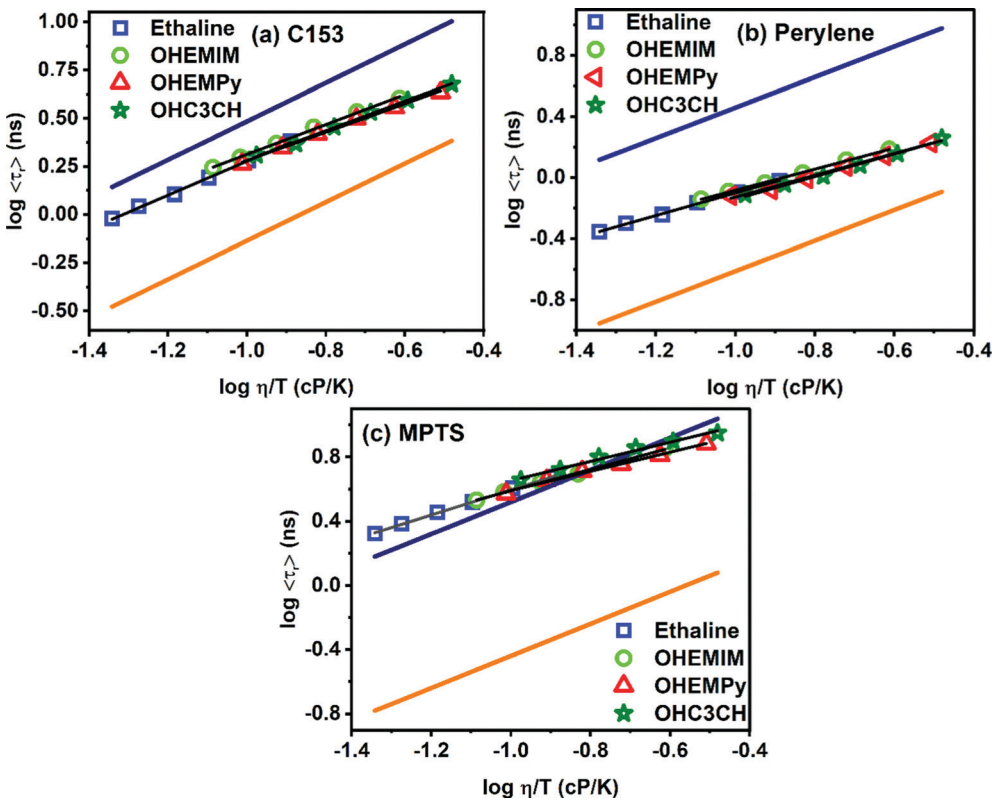


Fig. 6 Log–log plots of τ_r *versus* η/T of C153 (a), perylene (b) and MPTS (c) in both the DES and hydroxyl ILs. The solid orange and blue lines represent the slip and stick boundary conditions respectively. The solid black line represents a linear fit to the data points.

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Table 5 Parameters A and p obtained from the fits of $\log\langle\tau_r\rangle$ τ_r *vs.* $\log\langle\eta/T\rangle$

	C153		Perylene		MPTS	
System	A	p	A	p	A	p
Ethaline	14.4 ± 0.81	0.88 ± 0.01	4.37 ± 0.38	0.74 ± 0.04	23.4 ± 1.15	0.78 ± 0.02
[OHEMIM][NTf_2]	12.5 ± 0.79	0.77 ± 0.02	4.07 ± 0.42	0.70 ± 0.02	17.8 ± 1.03	0.66 ± 0.03
[OHEMPy][NTf_2]	10.2 ± 0.69	0.73 ± 0.01	3.89 ± 0.36	0.71 ± 0.01	15.5 ± 0.97	0.60 ± 0.02
[OHC3CH][NTf_2]	10.8 ± 0.73	0.75 ± 0.01	3.98 ± 0.31	0.73 ± 0.02	17.8 ± 1.02	0.60 ± 0.03

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cause a change in the hydrogen bonding and thus change the structure of the medium which would also be expected to cause a change in the C_{obs} value. From Table 3, it can be noticed that the average C_{obs} value of C153 in ethaline is found to be relatively higher as compared to the hydroxyl ILs. This indicates that the friction experienced by C153 in ethaline is comparatively more than those in the hydroxyl ILs. Furthermore, data from Table 3 reveal that C153 experiences more hindered rotation in [OHEMIM][NTf_2] as compared to the other two hydroxyl ILs. Thus, in the present study, the observation of different friction coefficients for the probe solute in the different media that are employed in the study indicates that the micro-viscosity felt by the solute is different, and all these results evidently reveal that the microscopic structural organisations of the hydroxyl ILs are different from each other as well as different from that of ethaline.

To corroborate the above outcomes, we have fitted the τ_r value of C153 in all solvents with the temperature reduced viscosity (η/T) through a power-law relationship given by eqn (9)

$$\tau_r = A(\eta/T)^p \quad (9)$$

where A and p are two constants. Though this relation is purely empirical, still it is often used in the studies of rotational diffusion of fluorophores in DESs and ILs, as it helps in gaining insight into the microscopic structural organisation of a solvent system in a simple and clear fashion.[27,45,51,69] For homogeneous solvent, the value of the exponent p is considered to be close to one and its deviation from unity reflects the departure from SED hydrodynamic behaviour (breakdown of SED hydrodynamic theory).[27,45,51,69] Previously, it has been demonstrated that the distributions of relaxation rates for molecules which reside in different local environments in organized media like ILs are responsible for dynamic heterogeneity in that medium.[44,69–71] We note here that Biswas and co-workers have demonstrated that decoupling of the rotational diffusion of a solute from the viscosity of the medium is caused by the dynamic heterogeneity of the medium.[44,50,51] Thus, the more the deviation of p from unity, the more the viscosity–diffusion decoupling and the more the dynamic heterogeneity of the medium. The values of A and p obtained from the empirical fitting are shown in Table 4. From Table 4, it can be noticed that for both the DES and hydroxyl ILs the p values significantly deviate from unity, indicating dynamic heterogeneity in these classes of systems. Previously, it has been demonstrated for hydroxyl ILs that the fractional viscosity dependence of τ_r arises because of non-Brownian motions such as orientational jumps and hydrogen bond fluctuations.[38,50,51] Furthermore, for DESs also hydrogen bond fluctuations and the consequent motion of the constituent particles have been considered as the primary cause for the observed fractional viscosity dependence of τ_r in DESs.[29,52] In light of this, the fractional viscosity dependence of τ_r for the current solvent systems can also be rationalized by considering the fluctuation of the hydrogen bonding network and the inertia driven motion which is different from normal Brownian motions.

3.2.2. Perylene. As stated earlier, perylene being apolar will locate itself in the apolar region of a given medium and is believed to probe the apolar domain of the medium.[26] The time resolved anisotropy decays of perylene (Fig. 5b) in all the solvents under study are found to be much faster as compared to the other two probes (C153 and MPTS). This indicates that perylene does not experience any strong solute–solvent interaction like MPTS (*vide infra*). From Table S3 (ESI†), we can see that at any particular temperature the τ_r value of perylene in ethaline is smaller than that in the hydroxyl ILs because of the low viscosity of the later. When these data are analysed together by SED theory, from the log–log plot of τ_r *versus* (η/T) (Fig. 6b) one can notice that the τ_r values of [OHC3CH][NTf_2] lie more close to the slip line as compared to ethaline, indicating relatively faster rotation of the solute in the hydroxyl ILs. This observation is also evident from the estimated C_{obs} value for perylene at a given temperature, where the C_{obs} values of perylene are found to be low for the hydroxyl ILs (0.19) than that found for the DES (0.26). Moreover, the p values (Table 5) obtained from the empirical relation (eqn (9)) for perylene in both the DES and ILs are found to be almost comparable (0.75–0.70). This is because of the fact that perylene does not involve itself in any strong solute–solvent interaction or specific hydrogen bonding interaction with the concerned solvents, and hence, similar extents of dynamic heterogeneity are experienced by perylene in these two classes of solvent systems.[26,27]

3.2.3. MPTS. The polar ionic region of a medium is monitored by using MPTS as the probe solute as it locates itself in the polar ionic region of the given medium.[26,31,52,61] The time resolved fluorescence anisotropy decay of MPTS (Fig. 5c) is found to be much slower than that from the other two solute probes (perylene and C153). This indicates that MPTS moves to the compact region, which is the ionic region of the medium.[26] Moreover, it has been shown previously for ILs that since MPTS has hydrogen bond acceptors such as the sulphite group (SO_3^-) it can participate in hydrogen bonding interaction with the acidic hydrogen of the ILs. Similarly, in the present study also, it can be expected that the sulphite group of MPTS can involve in hydrogen bonding interaction with the hydroxyl group of both the DES and hydroxyl ILs as well as with the HBD of

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ethaline. From Table 3, we can see that at 298 K the τ_r values for MPTS are nearly 5 times higher than those for perylene in the DES and hydroxyl ILs. This hindered rotation of MPTS in these media indicates that the solute–solvent interaction is significantly strong for MPTS in all these media than that for perylene. Analysis of the data in light of SED theory also reveals that the τ_r values of MPTS in both the DES and hydroxyl ILs lie close to the stick line, thereby indicating the hindered rotation of MPTS in these media. Interestingly, upon a closer look at the log–log plot of τ_r *versus* (η/T) (Fig. 6c) one can see that the rotational motion of MPTS in the DES is the slowest as compared to those in the hydroxyl ILs, and thus the τ_r values for MPTS in the DES were found to lie completely above the stick line. This behaviour is generally called super-stick behaviour in hydrodynamic terminology.[26,31,69] This super-stick behaviour can also be visualized by looking at the C_{obs} values (Table 3) of MPTS in these media. Previously, the super-stick behaviour of MPTS in self-organized media like ILs has been rationalized by considering the solute–solvent specific hydrogen bonding interaction.[26,69] Furthermore, it has also been reported that the hindered rotation of charged solutes in DESs is caused by the hydrogen bonding interaction between the charge solute and the constituents of DESs.[52] Considering these facts, it can be concluded that the hydrogen bonding interaction of MPTS with the DES and hydroxyl ILs is primarily responsible for the hindered rotation of the solute in the respective media. Furthermore, the p value (Table 5) obtained from the empirical relation (eqn (9)) is found to be much lower for the DES (0.78) as compared to the hydroxyl ILs (0.60). This observation indicates that the hydroxyl ILs are dynamically more heterogeneous as compared to the DES. The difference in the extent of dynamic heterogeneity between the DES and hydroxyl ILs in the present study perhaps also indicates the different rates of relaxation of the hydrogen bond fluctuation for the solute–solvent or solvent–solvent intermolecular interactions.[44,71,72]

3.3. Study of self-diffusion coefficients through PFG-NMR

The differences in microstructural organisation between the DES and hydroxyl ILs are further investigated using the pulse field gradient NMR (PFG-NMR) technique. The PFG-NMR technique, which is non-invasive and chemically selective, is found to be useful to estimate the self-diffusion coefficients of different species of a mixture simultaneously and can provide information regarding size, shape and intermolecular interaction between the constituent species.[35–39]

In the present study, the translational diffusion coefficient (D) values for both the DES and hydroxyl ILs have been estimated in the temperature range 298–323 K by fitting the NMR data through the Stejskal–Tanner equation.[55] The cationic parts of both the DES and hydroxyl ILs have been monitored exclusively during the measurement and the measured D values of both the DES and ILs are given in Table 6. From Fig. 7a, a gradual increase in the D value with increase in temperature for all concerned systems can be observed. This observation can be attributed to the decrease of the viscosity of the medium with increase in temperature (Table S3, ESI†). Moreover, one can observe from Table 6 that the mobility of the choline cation of the DES at any particular temperature is fastest compared to those of the cationic components of all the hydroxyl ILs. This behaviour is also consistent with the viscosity of the medium (Table S3, ESI†). In ethaline, one can also notice that the D value of the HBD (*i.e.*, ethylene glycol) is more as compared the choline cation, indicating its higher mobility. This is because of the smaller size of the former than the later. However, upon careful inspection it can be noticed that, upon going from ethaline to its IL analogue, [OHC3CH][NTf_2], the increase in viscosity is observed to be more than two times ($\sim$2.5 times), whereas the decrease in D is estimated to be less than two times ($\sim$1.75 times). This disproportionate change in D with bulk viscosity clearly reveals that bulk viscosity is not the sole factor which governs the mobility of cations of both the DES and hydroxyl ILs. To have a deeper insight into this behaviour, the NMR data have been analysed in light of the famous Stokes–Einstein equation ($D = kT/6\pi\eta r$), and by plotting D against T/η, the slopes ($=k/6\pi r$) for both the DES and hydroxyl ILs are calculated. With the slope being inversely proportional to the hydrodynamic radii of the cations, a steeper slope represents a larger hydrodynamic radius. From Fig. 7b, the order of slopes for various diffusing species is estimated to be (Ch^+) > [$OHEMPy^+$] > [$OHEMIM^+$] > [$OHC3CH^+$]. The steepest slope for the choline cation of the DES suggests that the hydrodynamic radius of the DES is larger as compared to the hydroxyl ILs. Further, to support our observation, we have also calculated the hydrodynamic radii (r) of the cationic species of all the concerned solvents through the SE equation at different temperatures and the data are provided in Fig. 7(c) and Table S4 (ESI†). From Fig. 7(c) and Table S4 (ESI†), one can also notice the larger hydrodynamic radius of the choline cation (2.44 Å) as compared to those of the cationic species of the hydroxyl ILs (1.75–1.55 Å). It is pertinent to mention in this context that D'Agostino *et al.*[36] while working on choline chloride based DESs reported a larger hydrodynamic radius for choline cations in ethaline medium. Moreover, in a separate study, they also reported significant association between the HBD and the hydroxyl group of the choline cation under anhydrous conditions.[35] Later on, Ludwig and coworkers[73] also reported that significant cation–cation interaction takes place in hydroxyl containing ILs as compared to that in non-hydroxyl ILs despite the cation–cation repulsion. Moreover, the effects of

Table 6 Estimated translational diffusion coefficients ($D \times 10^{-11}$ m^2 s^{-1}) of the cationic species of both the DES and hydroxyl ILs as well as the HBD of the DES

Temperature (K)	[$Choline^+$]/EG[a]	[$OHEMIM^+$]	[$OHEMPy^+$]	[$OHC3CH^+$]
298	2.32/3.43	1.72	1.34	1.43
303	2.94/4.24	2.26	1.73	1.83
308	3.57/5.32	2.88	2.16	2.3
313	4.36/6.44	3.6	2.64	2.84
318	5.12/7.80	4.42	3.23	3.53
323	6.3/8.92	5.32	3.83	4.35

[a] Diffusion coefficient of ethylene glycol.

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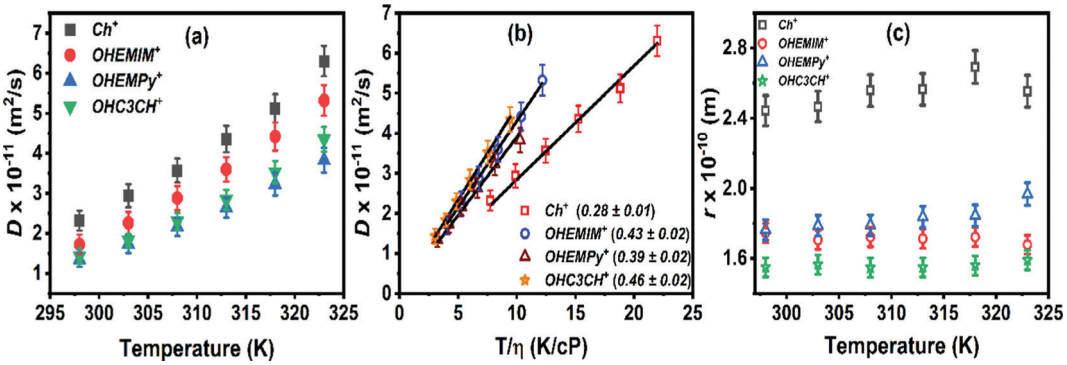


Fig. 7 (a) Temperature dependent variation of the self-diffusion coefficients of the DES and hydroxyl ILs. (b) *D vs. T*/η plots of the DES and hydroxyl ILs. The slope values (in J K^{-1} m^{-1}) for different solvents are shown in the legend. (c) Hydrodynamic radii of cationic species calculated through the SE equation.

cation–cation interaction on the hydrodynamic radii of hydroxyl containing ILs have also been studied by different groups[37] as well as by us[38] independently. It is interesting to note that in the present study despite the fact that the hydroxyl ILs can involve in intermolecular cation–cation H-bonding interaction, their hydrodynamic radii are found to be smaller as compared to that of ethaline. This observation probably indicates that because of the cation–cation repulsion, the intermolecular cation–cation interaction mediated by H-bonding interaction is somewhat weaker as compared to that of the intermolecular H-bonding interaction between the constituents of ethaline DES, and for this reason, the structure of ethaline is found to be more associated in nature as compared to those of the hydroxyl ILs, which makes the hydrodynamic radius of the former relatively larger than those of the later.

3.4. Fluorescence correlation spectroscopic (FCS) studies

FCS is a highly sensitive single molecule detection technique which is employed for the studies of different dynamical processes such as molecular translational diffusion, conformation fluctuations, reaction kinetics, and blinking dynamics.[74–76] Many researchers have also used FCS exclusively for the measurement of the translational diffusion of different probe molecules in DESs and RTILs, which provides information regarding the extent of solute–solvent association and the local environment of the medium.[29,52,74–78] Therefore, in the present investigation, FCS studies have also been performed to get a better understanding of the probes' diffusion dynamics and intermolecular solute–solvent interaction at a single molecular level.

Fig. 8 presents the fluorescence correlation decay curves ($G(\tau)$) of C153 and MPTS in ethaline medium. The data have been fitted through both the single component diffusion model and anomalous diffusion model (eqn (6)). However, looking at the fit residual (as well as reduced χ^2), it can be observed that the anomalous diffusion model fits the data (Fig. 8a and c) in a much better way as compared to the single component diffusion model (Fig. 8b and d). The normalized FCS decay curves of C153 and MPTS are provided in Fig. 8e and f respectively. The anomalous diffusion models fitting for C153 and MPTS in the hydroxyl ILs are provided in Fig. S7 and S8 (ESI†) respectively. The stretching exponent, β, of the anomalous diffusion model represents the extent of deviation from the single component diffusion model and gives a broad distribution of diffusion times (τ_D) of the probe molecules. The stretching exponent β values along with the translational diffusion coefficients (D_t) of the probes in their respective solvent systems estimated from eqn (6) and (7) are shown in Table 6.

From Table 7 one can notice that the diffusion coefficient (D_t) of C153 is found to be much higher in ethaline as compared to that in [OHC3CH][NTf_2] due to the fact that the medium viscosity of the former is lower than the later. Furthermore, the D_t value of the neutral solute C153 in a particular medium is found to be much higher than that of the ionic solute MPTS (Table 7). For example, the D_t value of C153 (24.8 μm^2 s^{-1}) in [OHC3CH][NTf_2] is estimated to be nearly 2 times larger than that of MPTS (6.23 μm^2 s^{-1}) in [OHC3CH][NTf_2]. This suggests that MPTS (with its sulphite group) can interact more strongly with the hydroxyl ILs as compared to that which can be observed for neutral C153. More interestingly, for the DES ethaline, the D_t value of C153 is found to be ~2.5 times higher than that for MPTS. This indicates that the frictional resistance to the translational diffusion of MPTS in ethaline is relatively more than those for MPTS in the hydroxyl ILs. Moreover, among the hydroxyl ILs, negligible differences in the D_t values of both probe molecules are observed in the case of both [OHEMPy][NTf_2] and [OHC3CH][NTf_2]. However, for [OHEMIM][NTf_2], even though the viscosity is found to be lower as compared to the other two hydroxyl ILs, the D_t value of MPTS remains almost the same in all the hydroxyl ILs. This observation is also suggestive of the fact that the extent solute–solvent interaction in [OHEMIM][NTf_2] is relatively stronger as compared to that in the other two hydroxyl ILs. Essentially all these results that are obtained from the translational diffusion dynamics of the probe solutes in both the DES and hydroxyl ILs nicely corroborate with the results obtained from time resolved anisotropy studies.

Apart from the translational diffusion coefficient, another parameter obtained from the fitting of FCS traces is the stretching exponent, β, whose values are found to lie between

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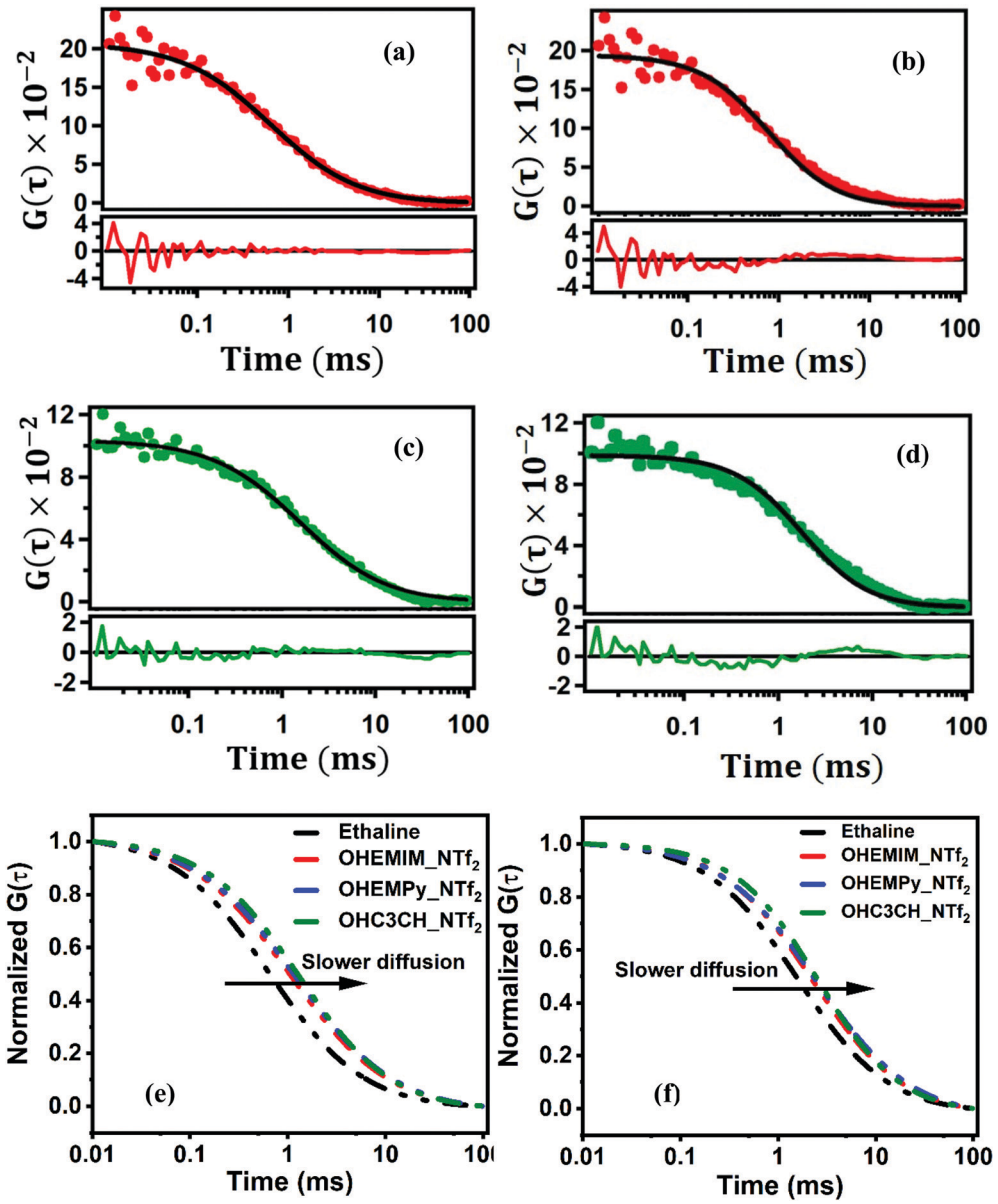


Fig. 8 Plots of $G(\tau)$ *vs.* time for C153 (a and b) and MPTS (c and d) in ethaline. Plots (a) and (c) are fitted through the anomalous diffusion model, while plots (b) and (d) are fitted through the single component diffusion model. (e) and (f) are the normalized FCS decay curves for C153 and MPTS, respectively, in both the DES and hydroxyl ILs.

Table 7 Estimated translational diffusion coefficients (D_t) along with the β values of C153 and MPTS in both the DES and hydroxyl ILs

	C153		MPTS	
System	D_t ($\mu m^2\ s^{-1}$)	β	D_t ($\mu m^2\ s^{-1}$)	β
Ethaline	24.8 ± 2.7	0.88 ± 0.02	10.0 ± 1.2	0.90 ± 0.03
[OHEMIM][NTf_2]	15.0 ± 1.8	0.84 ± 0.03	7.07 ± 0.9	0.85 ± 0.04
[OHEMPy][NTf_2]	13.4 ± 1.6	0.83 ± 0.03	6.48 ± 0.8	0.85 ± 0.03
[OHC3CH][NTf_2]	12.3 ± 1.7	0.85 ± 0.02	6.23 ± 0.8	0.86 ± 0.02

0.9 and 0.85 for both probes in all the solvent systems under study. The β values (which are well below unity) indicate that the diffusion dynamics of the solute molecules in these media are diverse in nature in terms of their translational diffusion time.[52,56] It has been well documented that for viscous heterogeneous media like DESs and RTILs, the mean square displacement (MSD) of solute molecules does not follow the normal Brownian diffusion ($\beta = 1$), instead it follows a sub-diffusive behaviour, $r(\tau)^2 \propto \tau^{\beta}$ (where $\beta < 1$).[56] As for conventional homogeneous solvents the diffusion of these solute molecules follows a single component diffusion model ($\beta = 1$), the deviation of the β value from unity indicates the presence of dynamic heterogeneity in all these media.[52,56] In the current study, as can be seen, the value of β deviates more for the hydroxyl ILs (*e.g.*, $\beta = 0.83$ for C153 in [OHEMPy][NTf_2]) as compared to the DES ($\beta = 0.88$ for C153 in ethaline), indicating more dynamic

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heterogeneity in the hydroxyl ILs as compared to the DES, which is also consistent with the result obtained from the time resolved anisotropy measurements.

4. Conclusion

The present study has been undertaken basically to understand the differences in microscopic behaviour in terms of intermolecular interactions, dynamics and structures between DESs and RTILs. For this purpose, a choline chloride-based DES (ethaline) and three hydroxyl functionalized ionic liquids have been employed so that, apart from the hydrogen bonding interaction, the roles of various other types of intermolecular interactions among the constituent species of the solvent systems in governing their behaviours are understood. All these investigations have unravelled several interesting findings. Excitation wavelength dependent emission spectra have indicated that the hydroxyl ILs employed in these studies are spatially more heterogeneous as compared to ethaline. EPR spectral measurements have revealed that the polarities of the hydroxyl ILs employed in this study are significantly high, whereas the polarity of ethaline is found to be close to those of aliphatic poly-hydroxy alcohols. This hyper-polarity of the hydroxyl ILs has been rationalized by considering the intramolecular synergistic effect that exists between the hydroxyl group of the cation and the corresponding anion. Investigations of the rotational and translational diffusion dynamics of selected probes have indicated that the extent of solute–solvent interaction is relatively higher in the DES in comparison to the hydroxyl ILs. Interestingly, diffusion studies have revealed that decoupling of solute dynamics with medium viscosity is found to be more for the hydroxyl ILs than that for ethaline, indicating that the extent of dynamic heterogeneity is more in the hydroxyl ILs than in ethaline. Measurements and analysis of self-diffusion coefficients through NMR have suggested that ethaline has a larger hydrodynamic radius than the hydroxyl ILs, indicating a more associated structure in ethaline. Essentially, all the results obtained from these investigations have demonstrated that despite having similar functional moieties, different intra/intermolecular interactions (such as solute–solvent interaction, specific hydrogen bonding interaction, and coulombic interaction) operating at the microscopic level are considerably different between the DES and hydroxyl ILs. This further highlights that all these interactions and energetics lead to a distribution of relaxation rates as well as spatial rearrangement which are different for the DES and hydroxyl ILs and are primarily responsible for observing the difference in the micro-heterogeneous behaviour between the DES and hydroxyl ILs. These differences in intermolecular interactions and micro-heterogeneous behaviour are manifested in the structure–property relationship for these solvent systems. Hence, the outcomes of the present study have revealed that consideration of DESs as a subclass of ILs may not be appropriate and DESs should be treated as a special class of solvent systems having its own identity.

Conflicts of interest

There are no conflicts to declare.

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